# FIITJEE – JEE (Main)

BATCH – LKTYR235A03

## Physics, Chemistry & Mathematics
## PHASE - I

**12.08.2023**

**Time Allotted: 3 Hours** **Maximum Marks: 300**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## Important Instructions

**Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.**

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections.**
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts**: **Part-A & C** in the OMR. Part-B of OMR to be left unused
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices ext. except the Admit Card inside the examination hall / room.

### B. Filling of OMR Sheet:

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with ***Blue/Black Ball Point Pen*** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.
4. **Do not fold or make any stray marks on the Answer Sheet.**

### C. Marking Scheme for All Two Parts:

**(i)** **Part-A (01-20)** contains Twenty (20) multiple choice objective questions which have four (4) options each and only one correct option. Each question carries **+4 marks** will be awarded for every correct answer and **-1 mark** will be deducted for every incorrect answer.

**(ii)** **Part-C (01-05)** contains Five (05) Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive). Each question carries **+4 marks** will be awarded for every correct answer and **NO MARKS** will be deducted for every incorrect answer.

**Name of the Candidate :**________________________________

**Batch :**________________ **Date of Examination :**________________

**Enrolment Number :**________________________________

# USEFUL DATA

## PHYSICS

Acceleration due to gravity : g = 10 $m/s^2$
Planck constant : h = 6.6 × $10^{-34}$ $J-s$
Charge of electron : e = 1.6 × $10^{-19}$ $C$
Mass of electron : $m_e = 9.1 \times 10^{-31} kg$
Permittivity of free space : $\varepsilon_0 = 8.85 \times 10^{-12} C^2 / N-m^2$
Density of water : $\rho_{water} = 10^3 kg/m^3$
Atmospheric pressure : $Pa = 10^5 N/m^2$
Gas constant : R = 8.314 J $K^{-1} mol^{-1}$

## CHEMISTRY

| | | |
|---|---|---|
| Gas Constant | R = | 8.314 J $K^{-1} mol^{-1}$ |
| | = | 0.0821 Lit atm $K^{-1} mol^{-1}$ |
| | = | 1.987 ≈ 2 Cal $K^{-1} mol^{-1}$ |
| Avogadro's Number $N_a$ | = | 6.023 × $10^{23}$ |
| Planck's constant h | = | 6.625 $\times 10^{-34}$ $J.s$ |
| | = | $6.625 \times 10^{-27}$ $erg.s$ |
| 1 Faraday | = | 96500 coulomb |
| 1 calorie | = | 4.2 joule |
| 1 amu | = | 1.66 $\times 10^{-27}$ $kg$ |
| 1 $eV$ | = | $1.6 \times 10^{-19}$ $J$ |

Atomic No: H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9, Ne = 10, Na = 11, Mg = 12, Si = 14, Al = 13, P = 15, S = 16, Cl = 17, Ar = 18, K = 19, Ca = 20, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Zn = 30, As = 33, Br = 35, Ag = 47, Sn = 50, I = 53, Xe = 54, Ba = 56, Pb = 82, U = 92.

Atomic masses: H = 1, He = 4, Li = 7, Be = 9, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Si = 28, Al = 27, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Co= 59, Ni = 58.7, Cu = 63.5, Zn = 65.4, As = 75, Br = 80, Ag = 108, Sn = 118.7, I = 127, Xe = 131, Ba = 137, Pb = 207, U = 238.

# Section – I Physics

## PART – A

**(Single Correct Answer Type)**

This part contain **20 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. $\left|\vec{A}\times\vec{B}\right|^2+\left|\vec{A}.\vec{B}\right|^2$ is equal to

(A) $\left(\vec{A}+\vec{B}\right)^2$ (B) $\left(\vec{A}-\vec{B}\right)^2$
(C) $A^2+B^2$ (D) $A^2B^2$

2. The area of a parallelogram formed from the vectors $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{B} = 3\hat{i} - 2\hat{j} + \hat{k}$ as adjacent sides is

(A) $8\sqrt{3}$ units (B) 64 units
(C) 96 units (D) $4\sqrt{6}$ units.

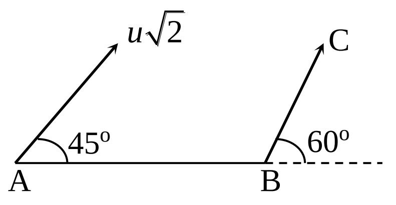
3. If $|\vec{A}| = 2$ and $|\vec{B}| = 4$, angle between them is $60^0$, $|\vec{A}-\vec{B}|$ is

(A) $\sqrt{13}$ (B) $3\sqrt{3}$
(C) $\sqrt{3}$ (D) $2\sqrt{3}$

4. A body is projected with a velocity u at an angle $\theta$ with the horizontal. The time after which its velocity vector is perpendicular to the initial velocity vector?

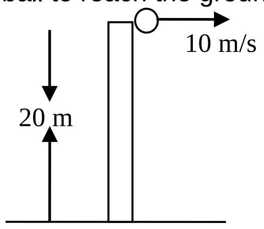
(A) $\frac{u\sin\theta}{g}$ (B) $\frac{g}{u\sin\theta}$
(C) $\frac{u}{g\sin\theta}$ (D) none of these

5. A particle is projected from a point A with velocity $u\sqrt{2}$ at an angle of 45° with horizontal as shown in figure. It strikes the plane BC at right angles. The velocity of the particle at the time of collision is



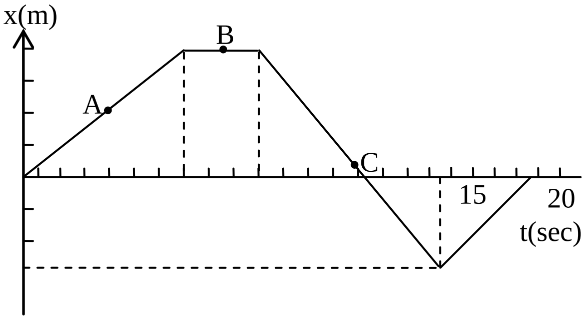

(A) $\frac{\sqrt{3}u}{2}$ (B) $\frac{u}{2}$
(C) $\frac{2u}{\sqrt{3}}$ (D) $u$

6. A ball is projected horizontally from the top of a tower of height 20 m with an initial speed 10 m/s. The time taken by the ball to reach the ground is (g = 10 m/s²)



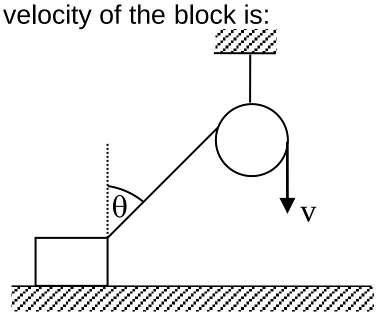

(A) 4 sec (B) 2 sec
(C) 1 sec (D) 1.5 sec

7. A person walks along an eastwest street, and a graph of his displacement from home is shown in figure. His average velocity for the whole time interval is –



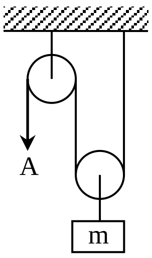

(A) 0
(B) 23 m/s
(C) 8.4 m/s
(D) None of the above

8. A block is dragged on a smooth plane with the help of a rope which is pulled by velocity v as shown in figure. The horizontal velocity of the block is:



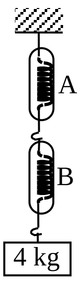

(A) v
(B) $v/\sin\theta$
(C) $v\sin\theta$
(D) $v/\cos\theta$

9. A load of mass M = 100 kg is supported is a vertical plane by a string and pulleys. If the tree end A of the string is pulled vertically downward with an acceleration a = 2 m/s$^2$, the tension in string is:



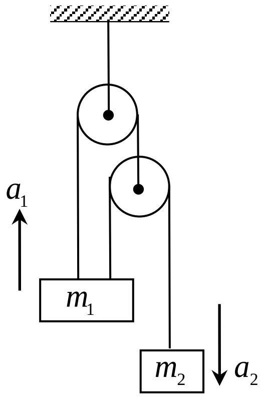

(A) 500 N
(B) 550 N
(C) 600 N
(D) 1000 N

10. A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will read respectively :



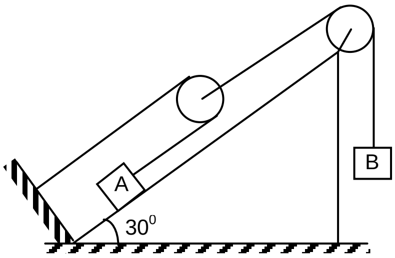

(A) 4 kg and zero kg
(B) zero kg and 4 kg
(C) 4 kg and 4 kg
(D) 2 kg and 2 kg

11. Find the relation between acceleration of two blocks shown in diagram.



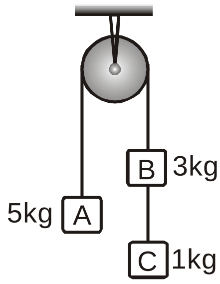

(A) $a_2 = 2a_1$
(B) $a_1 = 3a_2$
(C) $3a_1 = a_2$
(D) $a_1 = 2a_2$

12. In system shown in figure $m_B = 4kg$ and $m_A = 2kg$. The pulleys are massless and friction is absent everywhere. The acceleration of block A is $g = 10 m/s^2$



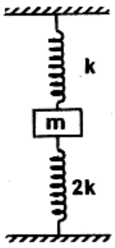

(A) $\frac{10}{3} m/s^2$
(B) $\frac{20}{3} m/s^2$
(C) $2\ m/s^2$
(D) 4 $m/s^2$

13. Three weights are hanging over a smooth fixed pulley as shown in the figure. What is the tension in the string connecting weights B and C?



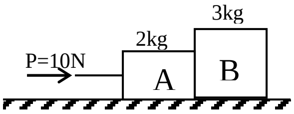

(A) g
(B) g/9
(C) 8g/9
(D) 10g/9

14. A block of mass m is connected to two springs of spring constant 2k and k respectively as shown in vertical plane. At equilibrium both spring are compressed by same length. If suddenly lower spring are compressed by same length. If suddenly lower spring is cut what is acceleration of block just after spring is cut.



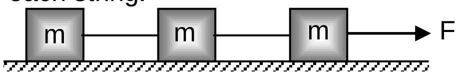

(A) 2g downward
(B) g downward
(C) g upward
(D) none of theses

15. Block A and B have masses of 2kg and 3kg respectively. The ground is smooth. P is an external force of 10 N. The force exerted by B on A is



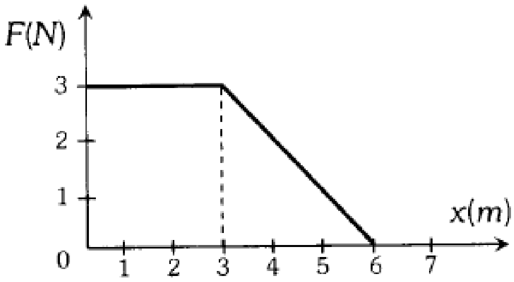

(A) 4N
(B) 6N
(C) 8 N
(D) 10 N

16. Three freight cars of mass m are tied with strings and are pulled with force F. Friction is negligible. The ratio of the forces in each string.



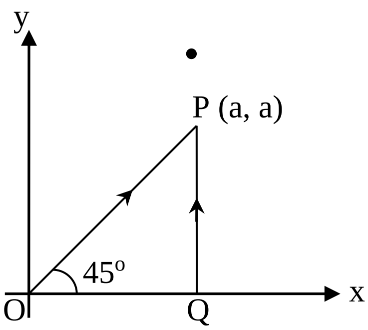

(A) 3 : 2 : 1
(B) 1 : 2 : 3
(C) 1 : 1 : 1
(D) 3 : 1 : 2

17. A force F acting on an object varies with distance $x$ as shown here. The force is in Newton and $x$ in metre. The work done by the force in moving the object from $x = 0$ to $x = 6$ m is




(A) 4.5 J
(B) 13.5 J
(C) 9.0 J
(D) 18.0 J

18. A particle is moved from (0, 0) to ($a$, $a$) under a force $\vec{F} = \left(3\hat{i} + 4\hat{j}\right)$ from two paths. Path 1 is OP and path 2 is OQP. Let $W_1$ and $W_2$ be the work done by this force in these two paths. Then

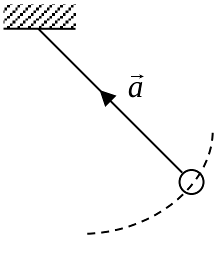



(A) $W_1 = W_2$
(B) $W_1 = 2W_2$
(C) $W_2 = 2W_1$
(D) $W_2 = 4W_1$

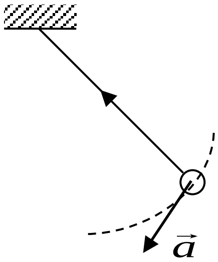
19. The potential energy as a function of the force between two atoms in a diatomic molecule is given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where $a$ and $b$ are positive constants and $x$ is the distance between the atoms. The position of stable equilibrium for the system of the two atoms is given by

(A) $x = \frac{a}{b}$
(B) $x = \sqrt{\frac{a}{b}}$
(C) $x = \frac{\sqrt{3a}}{b}$
(D) $x = \sqrt[6]{\left(\frac{2a}{b}\right)}$

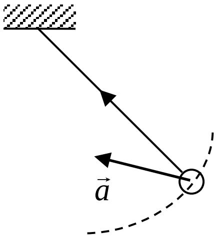
20. A sample pendulum is oscillating without damping. When the displacement of the bob is less than maximum, its acceleration vector $\vec{a}$ is correctly shown in



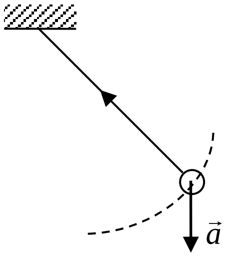

(A)




(B)




(C)



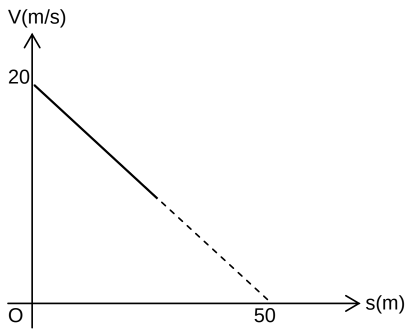

(D)

## PART – C

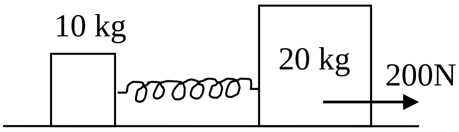
**(Integer Answer Type)**

This part contains **5 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9.

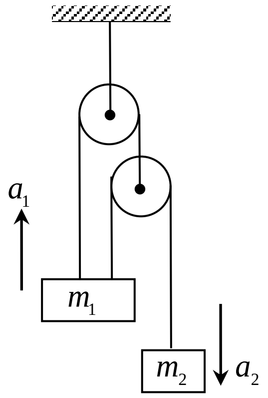
1. Referring to v –s diagram, find magnitude of acceleration of the particle when its velocity becomes half of the initial velocity



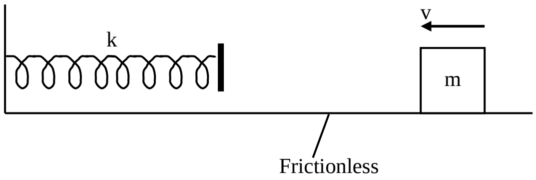

2. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in the figure. A force of 200 N acts on the 20 kg mass. At the instant shown the 10 kg mass has acceleration 12 m/s$^2$. What is the acceleration (in m/s$^2$) of 20 kg mass?



3. The relation between acceleration of two blocks shown in figure is $\frac{a_2}{a_1} = n$. Find *n*.




4. A block whose mass m is 5.7 kg slides on a horizontal frictionless tabletop with a constant speed v of 1.2 m/s. It is brought momentarily to rest as it compresses a spring in its path by distance 37/x cm. Find x. The spring constant k is 1500 N/m.




5. It is given that under a force $\vec{F} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ a body is displaced from position vector $\vec{r}_1 = (\hat{i} + \hat{j} + \hat{k})$ to a position vector $\vec{r}_2 = -4\hat{i} + 6\hat{j} + 2\hat{k}$, then work done in SI units is____________.

# Section – II Chemistry

## PART – A

**(Single Correct Answer Type)**

This part contain **20 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. In H-atom, if 'x' is the radius of the first Bohr orbit, de Broglie wavelength of an electron in 3rd orbit is :
(A) $3\pi x$ (B) $6\pi x$ (C) $\frac{9x}{2}$ (D) $\frac{x}{2}$

2. $Pb^{+4}$ is less stable than $Pb^{+2}$, because of
(A) inert pair effect (B) small size
(C) high ionisation potential (D) high electronegatively

3. Which of the following has least ionic character?
(A) $Cu_2Cl_2$ (B) KCl
(C) CsCl (D) $BaCl_2$

4. Bond length of which of the following types of bonds is maximum?
(A) $sp^2 - sp^2$ (B) $sp - sp$
(C) $sp^3 - sp^3$ (D) $sp^3 - sp^2$

5. The $IP_1$, $IP_2$, $IP_3$, $IP_4$, $IP_5$ of an element are 7.1, 14.3, 34.5, 46.7, 162.2 eV respectively. The valency of the element is
(A) 1 (B) 2
(C) 3 (D) 4

6. How many moles of electrons weigh one kilogram?
(Mass of electron = $9.108\times10^{-31}$ kg, Avogadro number = $6.022\times10^{23}$)
(A) $6.022\times10^{31}$ (B) $\frac{1}{9.108}\times10^{31}$
(C) $\frac{6.022}{9.108}\times10^{54}$ (D) $\frac{1}{9.108\times6.022}\times10^{8}$

7. The molecular weight of haemoglobin is about 65000 g/mol. Haemoglobin contains 0.35% Fe by mass. How many iron atoms are there in a haemoglobin molecule?
(A) 4 (B) 6
(C) 3 (D) 8

8. $Na^+$ is isoelectronic with
(A) $Ca^{2+}$ (B) $C^{4-}$
(C) $O^-$ (D) $K^+$

9. Which of the following process is accompanied with evolution or release of energy?
(A) $K(g)\longrightarrow K^+(g)+e^-$ (B) $Cl(g)+e^-\longrightarrow Cl^-(g)$
(C) $O^-(g)+e^-\longrightarrow O^{2-}(g)$ (D) $Mg^+(g)\longrightarrow Mg^{2+}(g)+e^-$

10. What is the azimuthal quantum number of the valence electron of potassium?
(A) Zero (B) One
(C) Two (D) Three

11. $\alpha$ is the angle between two lone pairs in $H_2O$, $\beta$ is the angle between any two lone pairs in $XeF_2$, $\gamma$ is the angle between two lone pairs in $ClF_3$. The correct order is

(A) $\gamma > \beta > \alpha$
(B) $\alpha > \beta > \gamma$
(C) $\beta > \gamma > \alpha$
(D) $\alpha > \gamma > \beta$

12. Which sequence correctly describes a relative bond strength, of oxygen molecule, superoxide ion and peroxide ion?

(A) $O_2 < O_2^- < O_2^{2-}$
(B) $O_2 > O_2^- > O_2^{2-}$
(C) $O_2 < O_2^- > O_2^{2-}$
(D) $O_2 > O_2^- < O_2^{2-}$

13. The most thermally stable sulphate is

(A) $BeSO_4$
(B) $BaSO_4$
(C) $MgSO_4$
(D) $CaSO_4$

14. Among LiCl, $BeCl_2$, $BCl_3$ and $CCl_4$, the covalent character follows the order

(A) LiCl < $BeCl_2$ > $BCl_3$ > $CCl_4$
(B) LiCl > $BeCl_2$ < $BCl_3$ < $CCl_4$
(C) LiCl < $BeCl_2$ < $BCl_3$ < $CCl_4$
(D) LiCl > $BeCl_2$ > $BCl_3$ > $CCl_4$

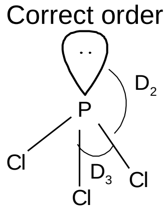
15. Which one of the following has highest bond angle?

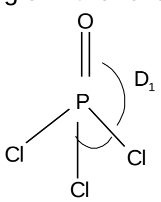
(A) $NH_3$
(B) $PH_3$
(C) $AsH_3$
(D) $SbH_3$

16. Which of the following is paramagnetic in nature?

(A) $C_2^{2-}$
(B) $N_2^{2+}$
(C) $O_2^+$
(D) $B_2^{2+}$

17. Correct order of bond angle in the following species it




(A) $D_1 > D_2 > D_3$
(B) $D_3 > D_2 > D_1$
(C) $D_1 > D_3 > D_2$
(D) $D_2 > D_1 > D_3$

18. A trigonal bipyrammidal complex is formed by hybridization of which atomic orbitals?

(A) $s, p_x, p_y, p_z, d_{x^2-y^2}$
(B) $s, p_x, p_y, d_{x^2-y^2}$
(C) $s, p_x, p_y, p_z, d_z^2$
(D) $s, p_x, p_y, p_z, d_{xy}$

19. The correct order of basic nature is

(A) LiOH < NaOH < KOH < RbOH < CsOH
(B) NaOH > KOH > RbOH > CsOH > LiOH
(C) LiOH < KOH < NaOH < RbOH < CsOH
(D) CsOH > RbOH > LiOH > KOH > NaOH

20. The hybridisation of atomic orbitals of nitrogen in $NO_2^+$, $NO_3^-$ and $NH_4^+$ are

(A) sp, $sp^3$ and $sp^2$ respectively
(B) sp, $sp^2$ and $sp^3$ respectively
(C) $sp^2$, sp and $sp^3$ respectively
(D) $sp^2$, $sp^3$ and sp respectively

## PART – C

### (Integer Answer Type)

This part contains **5 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9.

1. Caffeine has a molecular weight of 194. If it contains 28.9% by mass of Nitrogen, number of Nitrogen in one molecule of Caffeine is?

2. Number of moles of electrons present in 11.2 L of $NH_3$ at S.T.P. will be __________.

3. Ratio of potential energy to that of total energy of an electron in an hydrogen atom is equal to

4. How many of the following molecules are having this bond order = 2?
$N_2, N_2^-, N_2^{2-}, O_2, CN^+, C_2, CO, NO, F_2, B_2$

5. The observed dipole moment of HI is 0.38D. Calculate the percentage ionic character of the H–I bond if its bond length is 1.61 $\overset{o}{A}$.

# Section – III Mathematics

## PART – A
### (Single Correct Answer Type)

This part contain **20 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. If $y = \ln\left(\cos\sqrt{x}\right)$, then $\frac{dy}{dx}$ is equal to
(A) $-\frac{1}{2\sqrt{x}}\tan\sqrt{x}$
(B) $-\frac{1}{2\sqrt{x}}\cot\sqrt{x}$
(C) $-\frac{1}{2\sqrt{x}}\cos\sqrt{x}$
(D) None of these

2. $|x - 1| + |x + 1| < 1$ for
(A) $-1 < x < 1$
(B) $x > 1$ or $x < -1$
(C) $\{-1, 1\}$
(D) no value of x

3. $\log_{n^p} n^q$ is equal to
(A) $\frac{q}{p}$
(B) $\frac{p}{q}$
(C) pq
(D) $p^2$

4. Solution of $3^{x+2} > \left(\frac{1}{9}\right)^{1/x}$ is
(A) $x \in (0, \infty)$
(B) $x \in (-\infty, \infty)$
(C) $x \in (1, \infty)$
(D) none of these

5. Solution of $x^{\log_5 x} > 5$ is
(A) $x \in (2, \infty)$
(B) $x \in (0, 1/5) \cup (5, \infty)$
(C) $x \in (1, \infty)$
(D) $x \in (1, 5)$

6. Solution of $\log_{\log_5 x} 5 = 2$ is
(A) $x = \sqrt{5}$
(B) $x = 5^{\sqrt{5}}$
(C) $x = 5^5$
(D) none of these

7. The set of solution $\left|x^2 + x\right| = x^2 + x$ is given by
(A) $(-\infty, -1)$
(B) $[0, \infty)$
(C) $[-1, 0]$
(D) $(-\infty, -1] \cup [0, \infty)$

8. The set of real values of x such that $||(x - 1)| - 1| \le 1$ is
(A) $[-1, 3]$
(B) $[0, 2]$
(C) $[-1, 1]$
(D) none of these

9. If the mid point of join of $(x, y + 1)$ and $(x + 1, y + 2)$ is $\left(\frac{3}{2}, \frac{5}{2}\right)$, then the mid point of join of $(x - 1, y + 1)$ and $(x + 1, y - 1)$ is
(A) $(-1, -1)$
(B) $(-1, 1)$
(C) $(1, -1)$
(D) $(1, 1)$

10. Let A ={a, b, d, e}, B ={c, d, f, m}, C={a, l, m, o}, then $C \cap (A \cup B)$ will be given by
(A) {a, d, l, m}
(B) {b, c, l, o}
(C) {a, m}
(D) {a, b, c, d, f, l, m, o}

11. The least value of the expression $3x^2 + 2x + 2 \quad x \in R$ is
(A) $\frac{5}{3}$ at $x = -\frac{1}{3}$
(B) $\frac{2}{3}$ at $x = -\frac{1}{3}$
(C) $\frac{10}{3}$ at $x = -\frac{1}{3}$
(D) $\frac{10}{2}$ at $x = \frac{1}{3}$

12. If $\frac{(x-5)(|x+1|)}{(x^2+x+5)(x^2-4x-5)} > 0$, then x satisfies
(A) (–1, 5)
(B) (–1, ∞) – {5}
(C) [–1, ∞)
(D) (5, ∞)

13. If $\cos(A-B) = \frac{3}{5}$ and tanA tanB = 2, then
(A) $\cos A \cos B = \frac{1}{5}$
(B) $\sin A \sin B = -\frac{2}{5}$
(C) $\cos(A+B) = -\frac{2}{5}$
(D) $\sin A \sin B = \frac{4}{5}$

14. If $y = x^{x^2}$, then $\frac{dy}{dx}$ at x = e is
(A) $3e^{e^2+1}$
(B) e
(C) 1
(D) $3e^{e^2-1}$

15. $\left(2+\sqrt{3}\right)\sin\theta + 2\cos\theta$ lies between
(A) $-\sqrt{3}$ and $\sqrt{3}$
(B) $-\left(2+\sqrt{3}\right)$ and $\left(2+\sqrt{3}\right)$
(C) $-\left(2+\sqrt{5}\right)$ and $\left(2+\sqrt{5}\right)$
(D) none of these

16. If $\frac{|x+2|(x+3)(x-2)}{(x+1)} > 0$ then x belongs to
(A) (–3, –1) ∪ (2, ∞)
(B) (–∞, –3) ∪ (–1, 2)
(C) (–3, 2)
(D) (–1, ∞)

17. If the line y – mx + m –1 = 0 cuts the circle $x^2 + y^2 - 4x - 4y + 4 = 0$ at two real points, then m belongs to
(A) [1, 1]
(B) [-2, 2]
(C) (-∞, ∞)
(D) [-4, 4].

18. The equation of line, so that the mirror image of the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ in this line is $x^2 + y^2 + 2x + 2y + 1 = 0$, is
(A) x + y +1 = 0
(B) 8x + 6y –11 = 0
(C) 6 x + 8y –11 = 0
(D) 6x – 8y –11 = 0

19. If the tangent at the P on the circle $x^2 + y^2 + 2x + 2y = 7$ meets the straight line 3x – 4y = 15 at a point Q on the x-axis, then length of PQ is
(A) $3\sqrt{7}$
(B) $4\sqrt{7}$
(C) $2\sqrt{7}$
(D) $\sqrt{7}$

20. The coordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$ nearest to the origin is
(A) (3, 1)
(B) (18, 6)
(C) (15, 5)
(D) none of these

## PART – C

**(Integer Answer Type)**

This part contains **5 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9.

1. The number of solution of $\log_4(x-1) = \log_2(x-3)$ is

2. P (3, 1), Q (6, 5) and R (x, y) are three points such that the $\angle$RPQ is a right angle and the area of $\Delta$ RPQ = 7. Then the number of such point R is

3. If $\tan\alpha = \frac{p}{q}$ and $\tan\beta = \frac{q}{p}$, then $\cos(\alpha+\beta)$ is equal to

4. Angle between the lines ax + 2y + 1 = 0 and x –y + 1 = 0 is $45^0$ then value of a is

5. If line y + x = 2 do not intersect any member of circles $x^2 + y^2 - ax = 0$ at two distinct point where a is parameter, then maximum value of $\left|\frac{a+4}{\sqrt{2}}\right|$ is

# FIITJEE **INTERNAL TEST**

**BATCH: LKTYR235A03 (JEE Main)**

## PHASE TEST – I

## ANSWER KEY

### SECTION - I: PHYSICS

**PART - A**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | **D** | 2. | **D** | 3. | **D** | 4. | **C** |
| 5. | **C** | 6. | **B** | 7. | **A** | 8. | **B** |
| 9. | **B** | 10. | **C** | 11. | **C** | 12. | **A** |
| 13. | **D** | 14. | **A** | 15. | **B** | 16. | **A** |
| 17. | **B** | 18. | **A** | 19. | **D** | 20. | **C** |

**PART - C**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | **4** | 2. | **4** | 3. | **3** | 4. | **5** |
| 5. | **9** | | | | | | |

### SECTION - II: CHEMISTRY

**PART - A**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | **B** | 2. | **A** | **3.** | **A** | 4. | **C** |
| 5. | **D** | 6. | **D** | 7. | **A** | 8. | **B** |
| 9. | **B** | 10. | **A** | 11. | **A** | 12. | **B** |
| 13. | **B** | 14. | **C** | 15. | **A** | 16. | **C** |
| 17. | **D** | 18. | **C** | 19. | **A** | 20. | **B** |

**PART - C**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | **4** | 2. | **2** | 3. | **2** | 4. | **4** |
| 5. | **5** | | | | | | |

### SECTION - III: MATHEMATICS

**PART - A**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** | **A** | 2. | **D** | 3. | **A** | 4. | **A** |
| 5. | **B** | 6. | **B** | 7. | **D** | 8. | **A** |
| 9. | **D** | 10. | **C** | 11. | **A** | 12. | **B** |
| 13. | **A** | 14. | **A** | 15. | **D** | 16. | A |
| 17. | **C** | 18. | **C** | 19. | **C** | 20. | **A** |

**PART - C**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | **1** | 2. | **2** | 3. | **0** | 4. | **0** |
| 5. | **4** | | | | | | |